Reframing Bulk Refund Booking Detection as a Time-to-Event Problem with Risk Presentation

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Abstract

Bulk Refund Booking (BRB) occurs when a small number of users reserve many seats for personal purposes rather than actual ticket use and later refund most of them, creating artificial seat shortages and demand distortion. Prior work has shown that BRB can be proactively detected by separating risky and normal behavior with unsupervised pseudo-labeling and then training cumulative binary risk models. However, binary detection identifies whether a user is suspected of BRB, but not which flagged users should be reviewed first or when intervention is needed. This paper reframes BRB detection as a time-to-event risk problem that estimates both the likelihood of reaching a BRB-risk state and the remaining time until that transition. Using large-scale ticketing data from Korea's Super Rapid Train (SRT) system, we compare alternative refund-time definitions and identify the first refund that enters a BRB-risk state defined by unsupervised pseudo-labeling as an actionable event label. We further present a practitioner-oriented interface that visualizes full-period date-wise risk probabilities, expected event timing, and selected-date behavioral evidence. The results suggest that time-to-BRB modeling can support earlier prioritization, better intervention timing, and more interpretable operational review.

CCS Concepts

• **Human-centered computing** → **Visual analytics**; **Interactive systems and tools**; • **Computing methodologies** → **Machine learning**; • **Applied computing** → *Transportation.*

Keywords

Bulk Refund Booking, time-to-event prediction, behavioral risk modeling, ticketing systems, decision support

1 Introduction

Korea's Super Rapid Train (SRT) system [8], launched in 2016, now serves more than 25 million passengers annually [5, 10]. Its online and mobile reservation services improve convenience, but can also be misused through Bulk Refund Booking (BRB) [9], where a small number of users reserve many seats and later refund most of them, as illustrated in Figure 1. BRB may be driven by resale-oriented booking, adjacent-seat holding, reward accumulation, or schedule uncertainty [2], creating artificial seat shortages and demand distortion that can reach approximately 20% of daily demand [1, 6, 7]. These effects emerge not from a single fraudulent transaction, but from reservation and refund actions that accumulate over time and gradually turn into operational risk.

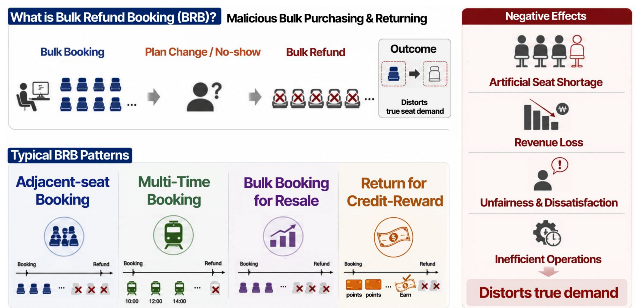

Figure 1: Illustrative overview of Bulk Refund Booking (BRB).

Prior work on BulkShield [7] addressed this sequential nature by using cumulative purchase and refund behavior to distinguish BRB-risk users from normal users, showing that BRB can be detected proactively at booking time. However, this formulation remains a binary risk detection task: it predicts whether a user is likely to be risky, but does not estimate when the user's behavior will reach an actionable BRB-risk state. This distinction matters operationally because large-scale ticketing services may flag many suspicious users, while practitioners cannot review or intervene in all cases with equal urgency.

Without event-time estimates, flagged users with similar risk scores may appear equally important even when their expected risk emergence differs substantially. Practitioners therefore lack a clear guideline for prioritizing intervention, especially when deciding whether a suspicious user requires immediate review or can be monitored later. BRB monitoring should thus move beyond a static normal/risky decision and estimate both the probability of BRB-risk emergence and the remaining time until that emergence.

This study reframes BRB detection as a full-period time-to-event risk problem that estimates how BRB probability evolves and when accumulated reservation-refund behavior reaches a BRB-risk state. Based on this framing, this paper makes the following contributions:

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- We introduce a full-period time-to-BRB formulation that converts BRB monitoring from a single-window normal/risky decision into an early-warning task that jointly considers date-wise event likelihood and remaining time to risk emergence.
- We operationalize the BRB event time by comparing the first refund, first BRB-risk entry refund, and last refund, showing that the first BRB-risk entry refund best captures an actionable point for proactive intervention.
- We design a practitioner-facing risk presentation that links full-timeline risk trajectories with selected-date behavioral summaries and transaction-level evidence, helping reviewers interpret why intervention may be needed at a specific time.

Together, these contributions move BRB analysis beyond post-hoc detection toward review prioritization, intervention timing, and evidence-based decision support.

1.1 Bulk Refund Booking Patterns and Sequential Risk Accumulation

Flexible online and mobile ticketing systems can be strategically exploited through BRB for different purposes, such as selectively refunding adjacent seats, partially using automated bulk bookings for resale, reserving multiple dates or time slots under schedule uncertainty, or accumulating credit-card rewards before refunding tickets. Despite these different motivations, BRB shares a common temporal structure: risk does not arise from a single isolated refund, but gradually emerges as booking-refund actions accumulate over time.

This sequential accumulation motivates a proactive temporal formulation that estimates not only whether a user is risky at inference time, but also how the risk trajectory evolves and when it is likely to reach an actionable BRB state.

1.2 Time-to-BRB Problem Formulation

Building on the sequential nature of BRB risk, we formulate proactive detection as a time-to-event prediction problem at each inference time t. For each user i, we observe the booking-refund history before t within a preceding W-day window:

$$\mathbf{s}_{i,<t}^{(W)} = \{\mathbf{e}_{i,1}, \mathbf{e}_{i,2}, \dots, \mathbf{e}_{i,T_i}\}, \tag{1}$$

where each event $\mathbf{e}_{i,j}$ represents a booking or refund action and T_i is the number of observed events in the window. This sequence captures the behavioral context available to the system before making an intervention decision.

Given this historical sequence, the model estimates both the probability that the user's behavior will transition into a BRB risk state and the expected time until that transition occurs:

$$(p_{i,t}, \hat{T}_{i,t}) = F_\theta(\mathbf{s}_{i,<t}^{(W)}), \quad \hat{E}_{i,t} = \mathbb{I}(p_{i,t} \geq \tau), \tag{2}$$

where $p_{i,t} \in [0, 1]$ denotes the predicted probability of reaching an actionable BRB state after inference time t, $\hat{T}_{i,t}$ denotes the predicted time remaining until BRB emergence, and $\hat{E}_{i,t}$ is the binary event prediction obtained by applying a threshold τ to the probability.

The ground-truth pair $(T_{i,t}, E_{i,t})$ is constructed from future observations: $E_{i,t}$ indicates whether an actionable BRB event occurs within the prediction horizon, and $T_{i,t}$ denotes either the time to the first event when $E_{i,t} = 1$ or the censoring time when $E_{i,t} = 0$. At test time, $\hat{E}_{i,t}$ is obtained from $p_{i,t}$ using a validation-selected threshold τ; event prediction is evaluated by comparing $\hat{E}_{i,t}$ with $E_{i,t}$, and timing accuracy by comparing $\hat{T}_{i,t}$ with $T_{i,t}$ for observed events.

This formulation differs from fixed-time binary classification because the output is not limited to a single normal/risky decision. Instead, repeated inference over time produces probability estimates $\{p_{i,t}\}$ and predicted event timings $\{\hat{T}_{i,t}\}$, which can be visualized as a continuous risk trajectory with predicted BRB timing. This enables practitioners to monitor how risk evolves and decide when proactive intervention is warranted.

2 Methodology

Building on the time-to-event formulation above, our methodology specifies how actionable BRB events are labeled and how the resulting predictions are presented for operational use. First, Sec. 3.1 defines the event time for time-to-BRB modeling by moving beyond binary BRB classification and identifying when a user's sequential refund trajectory first becomes risky enough for proactive intervention. This process converts each user sequence into a time-to-event label (T, E) for intervention-oriented risk modeling.

Second, Sec. 3.2 presents predicted BRB risk in an interpretable form for practitioners. This risk presentation combines temporal risk trajectories, 28-days behavioral inspection from a selected date, and transaction-level evidence that explains the major contributors to elevated risk.

2.1 Defining the Event Time for Time-to-BRB Modeling

We define time-to-BRB labels in two steps: first assigning GMM-based BRB risk pseudo-labels from post-hoc refund outcomes, and then extracting an actionable event time from ticket-level refund trajectories.

2.1.1 GMM-based BRB Risk Pseudo-label Construction. BRB is not directly observed as an explicit transaction-level ground-truth label, but is revealed post hoc through accumulated refund outcomes. We therefore follow the GMM-based pseudo-labeling procedure from prior work [7] and define BRB risk using refund count, refund rate, and refund amount.

For each inference time t, we define the booking observation window as

$$[t - W_{\text{opt}} + 1, t], \quad W_{\text{opt}} = 28 \text{ days}. \tag{3}$$

This window length follows prior work that selected the shortest observation window with high separation between BRB-risk and normal groups and low variability across repeated sampling [7]. For tickets booked within this window, we summarize their final refund outcomes as

$$\mathbf{r}_i(W) = (\text{count}_i(W), \text{rate}_i(W), \text{amount}_i(W)). \tag{4}$$

A GMM is fitted to these refund summary vectors, and the component with the highest median refund amount is defined as the BRB risk cluster. Each customer-window is assigned an `is_risk` label and an `end_date` corresponding to the inference time. Because this label indicates only whether the window eventually corresponds to

BRB risk, we reconstruct ticket-level refund trajectories to identify when the sequential refund process first reaches a BRB event state.

2.1.2 Event-time Extraction from Sequential Refund Trajectories. To recover the timing of BRB risk emergence, we reconstruct raw ticketing logs into ticket-level refund trajectories. For each ticket, we extract the booking time, refund time, refund amount, and refund status, and link these records to the corresponding GMM-labeled booking observation window.

For each GMM-labeled sample, we use `end_date` as the inference time t and select only tickets booked within the observation window $[t - W_{\text{opt}} + 1, t]$. Refunds of these tickets are included even when they occur after t, because the goal is to trace the post-hoc refund trajectory of tickets that were observable at inference time. We then sort the refund events chronologically and define three candidate event times:

$$\tau_{\text{cand}} \in \{\tau_{\text{first}}, \tau_{\text{cross}}, \tau_{\text{last}}\}, \tag{5}$$

where τ_{first} is the first refund time, τ_{cross} is the first refund time at which the cumulative refund rate and cumulative refund amount enter the GMM-defined BRB risk cluster, and τ_{last} is the last refund time. By comparing the gaps between these candidate event times and the inference time t, we determine which refund event time is most appropriate for defining BRB as a proactive time-to-event risk problem.

Finally, after selecting the event time τ_i^{event}, we construct the time-to-event label as

$$T_i = \tau_i^{event} - t_i, \tag{6}$$

where T_i denotes the remaining time from inference to the BRB event state. The event indicator E_i is set to 1 if the BRB event state is observed in the refund trajectory, and 0 otherwise, indicating that no BRB event is observed within the available refund trajectory. This formulation allows the model to estimate not only whether BRB risk will emerge, but also how soon intervention may be required.

2.2 Presenting Time-to-BRB Risk for Practitioner Intervention

Time-to-BRB predictions become operationally useful only when they are translated into evidence that practitioners can use for intervention decisions [3, 4]. Unlike BulkShield's fixed 28-days binary output, our interface combines full-timeline date-wise BRB occurrence probabilities with selected-date evidence analysis. Practitioners can first inspect when risk changes over the entire period, then select a specific date to review 28-days LLM-based interpretation, user behavior analysis, and transaction-level data evidence in a unified view.

2.2.1 Temporal Risk Trajectory for Overall Risk Monitoring. The first layer visualizes each user's BRB occurrence probability across the full timeline. Through this date-wise trajectory, practitioners can identify when risk begins to increase and how many days remain before the predicted risk exceeds the operational threshold. The expected threshold-crossing point is presented in a D-N early-warning format, providing temporal evidence for deciding whether intervention is needed before the final BRB event is completed.

2.2.2 Selected-Time Short-Term Behavioral and Transaction Evidence Analysis. Unlike the previous system, which provided only the fixed most recent 28-days data [7], our interface dynamically analyzes the past one month from a practitioner-selected date on the global risk trajectory. This selected-date view combines LLM-based interpretation, short-term user behavior summaries, and transaction-level data analysis, including attention-score-based top-k key transaction extraction and XAI interpretation tags from prior work. It therefore explains the accumulated booking-refund process behind a risk increase at a specific point in time and provides concrete evidence for intervention.

3 Results

This section presents an initial analysis of the proposed time-to-BRB formulation by addressing the following research questions:

- **RQ1.** Which refund-time event in the sequential refund process is most appropriate as a BRB event label that practitioners can proactively detect for early intervention? (Sec. 3.2)
- **RQ2.** How should the UI present date-wise BRB occurrence probabilities and expected event times so that practitioners can decide whether to intervene before the transition to BRB risk? (Sec. 3.3)

3.1 Dataset

3.1.1 Raw Ticketing Data. We use a large-scale SRT ticketing dataset containing ticket-level purchase and refund events for 6,756,549 customers who booked train journeys through mobile and online platforms between January 1 and December 31, 2024. Each record includes 148 attributes covering customer and booking information, train and journey details, and payment, cancellation, and refund records, such as anonymized customer identifiers, purchase timestamps, travel times, fares, and refund histories.

3.1.2 Preprocessed Data for Time-to-BRB Analysis. To construct reliable time-to-BRB labels, we retained users with at least one purchase and a usage period of 60 days or longer, and removed erroneous records with invalid departure or arrival timestamps or departure–arrival gaps exceeding two days. After preprocessing, the dataset contained 1,697,500 users, with an average of 11.45 active days and 33.56 events per user. For each user, we randomly selected a booking inference time, applied the preceding W_{opt}-day observation window, and performed GMM-based pseudo-labeling. The analysis focuses on 40,740 sampled windows identified as BRB-risk cases, corresponding to 2.4% of the preprocessed user samples.

3.2 Time-to-BRB Labeling Case Analysis

We compare three candidate refund-time events in the sequential refund process: the first refund, the first BRB-risk entry refund, and the last refund. For each GMM-labeled risk window, we measure the time gap between the inference time and each candidate event time, as shown in Figure 2. The mean±std time gaps are -13.080±9.374, 7.514±7.824, and 11.138±9.450 days for the first refund, first BRB-risk entry refund, and last refund, respectively. Pairwise Wilcoxon signed-rank tests show significant differences among all three candidates [11].

Figure 2: Time gaps from inference to three candidate BRB event times. The shaded post-inference region indicates the actionable window for proactive intervention. Among the three candidates, the first BRB-risk entry refund best balances post-inference actionability and timely risk emergence.

This comparison allows us to examine whether each refund-time definition provides an operationally meaningful event label for proactive intervention, rather than simply identifying whether a user-window eventually becomes risky.

The results show that the first refund occurs before inference on average, suggesting that it reflects an early and isolated refund signal rather than the emergence of a BRB risk state. In contrast, the last refund occurs substantially later, making it closer to a post-hoc completion point than an actionable intervention target. The first BRB-risk entry refund provides a more appropriate event label because it marks the first point at which the accumulated refund trajectory enters the GMM-defined BRB risk state, while still occurring after inference with sufficient lead time for operational response. Therefore, we use the first BRB-risk entry refund as the BRB event time for constructing time-to-event labels.

3.3 Representative User Case Study for Time-to-BRB Risk Presentation

This section examines how the proposed time-to-BRB risk presentation supports practitioner intervention decisions using a representative adjacent-seat reservation and refund case. The selected user repeatedly secured adjacent or duplicate seats for weekend or holiday trips and later refunded part of those reservations, illustrating a BRB pattern that gradually forms through accumulated booking-refund behavior rather than a single transaction.

Figure 3 illustrates a review scenario in which a practitioner inspects the representative user's data on September 24, 2023. The predicted risk probability remains low during ordinary periods, rises around weekend travel dates, and exceeds the operational threshold during major Korean holidays such as Seollal (Korean Lunar New Year) and Chuseok (Korean Thanksgiving). By presenting this transition in a D-N early-warning format relative to the review date, the interface helps practitioners move beyond a single risky/normal decision and identify when intervention should be considered. This case illustrates how date-wise risk probabilities and expected event timing can be presented as an early-warning trajectory to support intervention decisions before the transition to BRB risk.

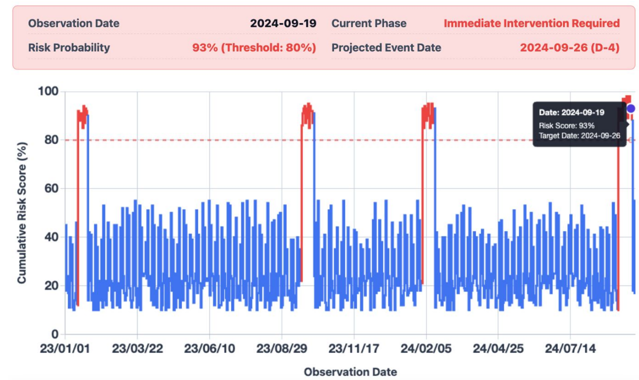

Figure 3: Time-to-BRB risk presentation interface for a representative adjacent-seat reservation and refund case.

4 Discussion and Future Directions

Our analysis shows that event-time definition is a central design decision for proactive BRB detection. The first refund can be too early and ambiguous, whereas the last refund often captures the pattern only after most harmful effects have occurred. The first BRB-risk entry refund provides a more actionable middle point by marking when accumulated refund behavior first enters a GMM-defined risk state, while still leaving time for review and response.

This event definition also affects how model outputs should be presented. A binary label can indicate risk, but it does not show urgency, progression, or the behaviors behind the increase. The proposed time-to-BRB presentation addresses this limitation by showing the full risk trajectory and then allowing practitioners to drill down into a selected high-risk date. There, the interface connects recent behavioral context with LLM-based interpretation and transaction-level evidence, helping practitioners decide both when intervention is needed and why that timing is justified.

This paper focuses on event-label construction and risk presentation rather than a complete survival model. The representative case study demonstrates the interface's potential, but broader practitioner evaluation is needed to assess usability, trust, workload reduction, and decision quality. Future work will develop survival-oriented BRB prediction models and evaluate them with operational SRT data.

5 Conclusion

This study reframed Bulk Refund Booking (BRB) detection from static risk classification into a time-to-event risk problem. We defined the first BRB-risk entry refund as an actionable event label and presented time-to-BRB risk using date-wise risk trajectories, selected-date behavioral summaries, and transaction-level evidence. By connecting intervention timing with behavioral rationale, the proposed approach provides a basis for more timely and interpretable BRB early-warning systems.

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